Intro to Probability T# Lecture Notes — Probability & Statistics

Module 1 — Probability & Counting

Lecture 1 — Basic Counting Rule

Core Idea

Counting = breaking a problem into stages.

Basic Counting Rule (Multiplication Rule)

If an experiment has r stages and each stage has n_i outcomes:

$$\text{Total outcomes} = n_1 \cdot n_2 \cdots n_r$$

Key Insight

- Multiply choices across stages
- Order of stages matters

Lecture 2 — Permutations & Combinations

Permutations (Order Matters)

$$P(n, r) = \frac{n!}{(n-r)!}$$

- Used when order matters (rankings, assignments)

Combinations (Order Does NOT Matter)

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- Used when order does not matter (groups, committees)

Key Idea

- Permutations → order matters
- Combinations → order does not matter

Lecture 3 — Multinomial Coefficient

Multinomial Coefficient

$$\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

Use Case

- Dividing items into groups of fixed sizes

Key Idea

- Generalization of combinations

Lecture 4 — Sample Space & Events

Definitions

- Sample space$S$: all possible outcomes
- Event$A \subseteq S$: subset of outcomes

Event Operations

Union:

$$A \cup B$$

Intersection:

$$A \cap B$$

Complement:

$$A^c$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Probability (Equally Likely Outcomes)

$$P(A) = \frac{|A|}{|S|}$$

Lecture 5 — Axioms of Probability

Axioms

1. $$0 \leq P(A) \leq 1$$

2. $$P(S) = 1$$

3. $$P\left(\bigcup A_i\right) = \sum P(A_i)$$

Key Formulas

Complement:

$$P(A^c) = 1 - P(A)$$

Union:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Lecture 6 — Practice Concepts

Core Idea

Combine counting + probability

Common Patterns

At least one:

$$P(\text{at least one}) = 1 - P(\text{none})$$

Birthday-type problems:

- Use complement
- Multiply sequential probabilities

Key Takeaways

- Break into stages
- Check if order matters
- Use complement when easier

Module 2 — Conditional Probability

Lecture 1 — Conditional Probability

Core Idea

Update probability with new information.

Definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

- Prior: $P(A)$
- Posterior: $P(A|B)$

Key Insight

- Restrict to event B
- Compute probability within that space

Lecture 2 — Multiplication Rule & Total Probability

Multiplication Rule

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

General Form

$$P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)\cdots$$

Law of Total Probability

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

Key Insight

- Break into cases
- Use trees

Lecture 3 — Bayes' Theorem

Bayes' Rule

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)}$$

Key Idea

- Reverse conditioning

Lecture 4 — Independence

Definition

$$P(A \cap B) = P(A)P(B)$$

Equivalent:

$$P(A|B) = P(A)$$

Key Differences

- Independent → no effect
- Disjoint → cannot happen together

Conditional Independence

$$P(A \cap B|E) = P(A|E)P(B|E)$$

Key Takeaways

- Update probabilities with info
- Multiply for joint events
- Use total probability for cases
- Bayes reverses conditioning
- Independence = no influence

Module 3 — Discrete Random Variables

Lecture 1 — Random Variables & PMF

Random Variable

A random variable is a function:

$$X : S \to \mathbb{R}$$

- Maps outcomes → numbers
- Events look like:$X = x$

Discrete Random Variable

- Takes finite or countable values
- Support = values where$P(X = x) > 0$

PMF (Probability Mass Function)

$$p_X(x) = P(X = x)$$

Valid PMF Conditions

-$p_X(x) \geq 0$
-$\sum_x p_X(x) = 1$

Lecture 2 — CDF & Functions of RVs

CDF

$$F_X(x) = P(X \leq x)$$

Key Relationship

- PMF → point probabilities
- CDF → accumulated probabilities

Function of Random Variable

If $Y = g(X)$:

$$P(Y = y) = \sum_{x:g(x)=y} P(X = x)$$

Key Insight

- Transform values, then recompute probabilities

Lecture 3 — Expectation & Variance

Expectation

$$E(X) = \sum_x x\, p_X(x)$$

- Weighted average

Function Expectation

$$E[g(X)] = \sum_x g(x) p_X(x)$$

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\mathrm{Var}(X) = E[(X - E(X))^2]$$

Alternative:

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Standard Deviation

$$SD(X) = \sqrt{\mathrm{Var}(X)}$$

Lecture 4 — Independence, Bernoulli, Binomial

Independence (Random Variables)

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

Bernoulli Distribution

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

- One trial (success/failure)

Properties

$$E(X) = p, \quad \mathrm{Var}(X) = p(1-p)$$

Binomial Distribution

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

-$n$trials

- Independent
- Same probability$p$

Lecture 5 — Binomial (Applications)

Conditions

- Fixed$n$
- Independent trials
- Same$p$
- Two outcomes

Properties

$$E(X) = np, \quad \mathrm{Var}(X) = np(1-p)$$

Key Trick

Use complement for “at least”:

$$P(X \geq k) = 1 - P(X < k)$$

Lecture 6 — Poisson & Negative Binomial

Poisson Distribution

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

- Counts events in interval
- Models rare events

Properties

$$E(X) = \lambda, \quad \mathrm{Var}(X) = \lambda$$

Approximation

$$\mathrm{Bin}(n, p) \approx \mathrm{Pois}(\lambda = np)$$

Negative Binomial

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

- Number of trials until r successes

Lecture 7 — Geometric & Hypergeometric

Geometric Distribution

$$P(X = x) = (1-p)^{x-1} p$$

- Trials until first success

Properties

$$E(X) = \frac{1}{p}, \quad \mathrm{Var}(X) = \frac{1-p}{p^2}$$

Hypergeometric Distribution

$$P(X = x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$$

- Sampling without replacement

Key Difference

- Binomial → independent trials
- Hypergeometric → dependent (no replacement)

Lecture 8 — Distribution Summary

When to Use

- Bernoulli → one trial
- Binomial → fixed number of trials
- Poisson → count over time/space
- Negative Binomial → until r successes
- Geometric → until first success
- Hypergeometric → sampling without replacement

Key Relationships

- Bernoulli = Binomial with $n = 1$

- Geometric = Negative Binomial with $r = 1$
- Binomial → Poisson (large n, small p)

Core Takeaways

- Random variables map outcomes to numbers
- PMF describes probabilities
- Expectation = average
- Variance = spread
- Choose distribution based on setup

Module 4 — Continuous Random Variables

Lecture 1 — Continuous RV, PDF & CDF

Continuous Random Variable

- Takes values over intervals
- Examples: height, time, distance

CDF

$$F_X(x) = P(X \leq x)$$

PDF

$$f(x) = \frac{d}{dx} F_X(x)$$

Probability via PDF

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Key Properties

-$f(x) \geq 0$
-$\int_{-\infty}^{\infty} f(x)\,dx = 1$
-$P(X = a) = 0$

Lecture 2 — Transformations & Expectation

Transformation

If $Y = g(X)$(monotonic):

$$f_Y(y) = f_X(g^{-1}(y)) \;\; \frac{d}{dy} g^{-1}(y)$$

Expectation (Continuous)

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$$

Function Expectation

$$E[g(X)] = \int g(x) f(x)\,dx$$

Key Idea

- Replace sums → integrals

Lecture 3 — Variance (Continuous)

Variance

$$\mathrm{Var}(X) = E[(X - E(X))^2]$$

Alternative:

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Key Insight

- Same formulas as discrete, but integrals

Lecture 4 — Uniform & Exponential

Uniform Distribution

$$f(x) = \frac{1}{b-a}, \quad a < x < b$$

- All values equally likely

Properties

$$E(X) = \frac{a+b}{2}, \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Exponential Distribution

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Properties

$$E(X) = \frac{1}{\lambda}, \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property

$$P(X \geq s + t \mid X \geq s) = P(X \geq t)$$

Lecture 5 — Normal Distribution

PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Properties

- Mean:μ
- Variance:σ^2
- Symmetric, bell-shaped

Standardization

$$Z = \frac{X - \mu}{\sigma}$$

Lecture 6 — Empirical Rule & Chi-Square

Empirical Rule

-$P(|Z| < 1) \approx 0.68$
-$P(|Z| < 2) \approx 0.95$
-$P(|Z| < 3) \approx 0.997$

Chi-Squared Distribution

$$Q = Z_1^2 + \cdots + Z_k^2 \sim \chi_k^2$$

Key Idea

- Sum of squared standard normals

Lecture 7 — Gamma & Beta

Gamma Distribution

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$$

- Models waiting time

Key Relationship

$\mathrm{Gamma}(1, \lambda) = \mathrm{Exponential}$

Beta Distribution

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1$$

- Models proportions

Lecture 8 — Weibull & Log-Normal

Weibull Distribution

$$f(x) = \gamma \lambda x^{\gamma-1} e^{-\lambda x^\gamma}$$

- Flexible lifetime model

Log-Normal Distribution

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Key Idea

- If $\ln X \sim N(\mu, \sigma^2)$, then X is log-normal

Lecture 9 — Summary & Relationships

When to Use

- Uniform → equal likelihood
- Normal → symmetric data
- Exponential → waiting time
- Gamma → sum of waiting times
- Beta → proportions
- Chi-square → sum of squares
- Weibull → lifetime modeling
- Log-normal → positive, skewed data

Key Relationships

- Exponential ↔ Poisson (waiting vs count)
- Gamma → sum of exponentials
- Beta from Gamma ratios
- Normal → basis for many distributions

Core Takeaways

- Continuous → use integrals
- PDF gives density, not point probability
- CDF accumulates probability
- Choose distribution based on context

Module 5 — Moment Generating Functions

Lecture 1 — Moments & MGF

Moments

- k-th moment:

$$E(X^k)$$

- k-th central moment:

$$E[(X-\mu)^k]$$

Skewness

$$\text{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$

Kurtosis

$$\text{Kurt}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] - 3$$

Moment Generating Function (MGF)

$$M_X(t) = E(e^{tX})$$

Key Properties

-$M(0) = 1$

- Encodes all moments
- Must be finite near$t = 0$

Lecture 2 — Using MGF

Moments from MGF

- Take derivatives:

$$E(X^k) = M_X^{(k)}(0)$$

First Two Moments

$$E(X) = M_X'(0)$$

$$E(X^2) = M_X''(0)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Transformation Property

If $Y = a + bX$:

$$M_Y(t) = e^{at} M_X(bt)$$

Key Idea

- MGF turns calculus → moments

Lecture 3 — Sums of Random Variables

Sum of Independent RVs

If $U = X_1 + \cdots + X_n$:

$$M_U(t) = M_{X_1}(t) \cdots M_{X_n}(t)$$

Key Insight

- MGFs multiply for independent sums

Important Results

- Sum of normals → normal
- Sum of Poissons → Poisson

Example Pattern

If $X_i \sim \mathrm{Bern}(p)$, then:

$$\sum X_i \sim \mathrm{Binomial}(n, p)$$

Core Takeaways

- MGF uniquely determines distribution
- Derivatives give moments
- Products handle sums

- Useful for proving distribution resultsheory

Module 1 — Probability & Counting

Lecture 1 — Basic Counting Rule

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Use Case

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Probability (Equally Likely Outcomes)

$$P(A) = \frac{|A|}{|S|}$$

Lecture 5 — Axioms of Probability

Axioms

1. $$0 \leq P(A) \leq 1$$

2. $$P(S) = 1$$

3. $$P\left(\bigcup A_i\right) = \sum P(A_i)$$

Key Formulas

Complement:

$$P(A^c) = 1 - P(A)$$

Union:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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Core Idea

Combine counting + probability

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At least one:

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Birthday-type problems:

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Key Takeaways

- Break into stages
- Check if order matters
- Use complement when easier

Module 2 — Conditional Probability

Lecture 1 — Conditional Probability

Core Idea

Update probability with new information.

Definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

- Prior:$P(A)$
- Posterior:$P(A|B)$

Key Insight

- Restrict to event$B$
- Compute probability within that space

Lecture 2 — Multiplication Rule & Total Probability

Multiplication Rule

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

General Form

$$P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)\cdots$$

Law of Total Probability

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

Key Insight

- Break into cases
- Use trees

Lecture 3 — Bayes' Theorem

Bayes' Rule

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)}$$

Key Idea

- Reverse conditioning

Lecture 4 — Independence

Definition

$$P(A \cap B) = P(A)P(B)$$

Equivalent:

$$P(A|B) = P(A)$$

Key Differences

- Independent → no effect
- Disjoint → cannot happen together

Conditional Independence

$$P(A \cap B|E) = P(A|E)P(B|E)$$

Key Takeaways

- Update probabilities with info
- Multiply for joint events
- Use total probability for cases
- Bayes reverses conditioning
- Independence = no influence

Module 3 — Discrete Random Variables

Lecture 1 — Random Variables & PMF

Random Variable

A random variable is a function:

$$X : S \to \mathbb{R}$$

- Maps outcomes → numbers
- Events look like:$X = x$

Discrete Random Variable

- Takes finite or countable values
- Support = values where$P(X = x) > 0$

PMF (Probability Mass Function)

$$p_X(x) = P(X = x)$$

Valid PMF Conditions

-$p_X(x) \geq 0$
-$\sum_x p_X(x) = 1$

Lecture 2 — CDF & Functions of RVs

CDF

$$F_X(x) = P(X \leq x)$$

Key Relationship

- PMF → point probabilities
- CDF → accumulated probabilities

Function of Random Variable

If $Y = g(X)$:

$$P(Y = y) = \sum_{x:g(x)=y} P(X = x)$$

Key Insight

- Transform values, then recompute probabilities

Lecture 3 — Expectation & Variance

Expectation

$$E(X) = \sum_x x\, p_X(x)$$

- Weighted average

Function Expectation

$$E[g(X)] = \sum_x g(x) p_X(x)$$

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

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Alternative:

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Standard Deviation

$$SD(X) = \sqrt{\mathrm{Var}(X)}$$

Lecture 4 — Independence, Bernoulli, Binomial

Independence (Random Variables)

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

Bernoulli Distribution

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

- One trial (success/failure)

Properties

$$E(X) = p, \quad \mathrm{Var}(X) = p(1-p)$$

Binomial Distribution

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

-$n$trials

- Independent
- Same probability$p$

Lecture 5 — Binomial (Applications)

Conditions

- Fixed$n$
- Independent trials
- Same$p$
- Two outcomes

Properties

$$E(X) = np, \quad \mathrm{Var}(X) = np(1-p)$$

Key Trick

Use complement for “at least”:

$$P(X \geq k) = 1 - P(X < k)$$

Lecture 6 — Poisson & Negative Binomial

Poisson Distribution

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

- Counts events in interval
- Models rare events

Properties

$$E(X) = \lambda, \quad \mathrm{Var}(X) = \lambda$$

Approximation

$$\mathrm{Bin}(n, p) \approx \mathrm{Pois}(\lambda = np)$$

Negative Binomial

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

- Number of trials until r successes

Lecture 7 — Geometric & Hypergeometric

Geometric Distribution

$$P(X = x) = (1-p)^{x-1} p$$

- Trials until first success

Properties

$$E(X) = \frac{1}{p}, \quad \mathrm{Var}(X) = \frac{1-p}{p^2}$$

Hypergeometric Distribution

$$P(X = x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$$

- Sampling without replacement

Key Difference

- Binomial → independent trials
- Hypergeometric → dependent (no replacement)

Lecture 8 — Distribution Summary

When to Use

- Bernoulli → one trial
- Binomial → fixed number of trials
- Poisson → count over time/space
- Negative Binomial → until r successes
- Geometric → until first success
- Hypergeometric → sampling without replacement

Key Relationships

- Bernoulli = Binomial with $n = 1$
- Geometric = Negative Binomial with $r = 1$
- Binomial → Poisson (large n, small p)

Core Takeaways

- Random variables map outcomes to numbers
- PMF describes probabilities
- Expectation = average
- Variance = spread
- Choose distribution based on setup

Module 4 — Continuous Random Variables

Lecture 1 — Continuous RV, PDF & CDF

Continuous Random Variable

- Takes values over intervals
- Examples: height, time, distance

CDF

$$F_X(x) = P(X \leq x)$$

PDF

$$f(x) = \frac{d}{dx} F_X(x)$$

Probability via PDF

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

Key Properties

-$f(x) \geq 0$
-$\int_{-\infty}^{\infty} f(x)\,dx = 1$
-$P(X = a) = 0$

Lecture 2 — Transformations & Expectation

Transformation

If $Y = g(X)$(monotonic):

$$f_Y(y) = f_X(g^{-1}(y)) \;\; \frac{d}{dy} g^{-1}(y)$$

Expectation (Continuous)

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$$

Function Expectation

$$E[g(X)] = \int g(x) f(x)\,dx$$

Key Idea

- Replace sums → integrals

Lecture 3 — Variance (Continuous)

Variance

$$\mathrm{Var}(X) = E[(X - E(X))^2]$$

Alternative:

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Key Insight

- Same formulas as discrete, but integrals

Lecture 4 — Uniform & Exponential

Uniform Distribution

$$f(x) = \frac{1}{b-a}, \quad a < x < b$$

- All values equally likely

Properties

$$E(X) = \frac{a+b}{2}, \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Exponential Distribution

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Properties

$$E(X) = \frac{1}{\lambda}, \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property

$$P(X \geq s + t \mid X \geq s) = P(X \geq t)$$

Lecture 5 — Normal Distribution

PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Properties

- Mean:μ
- Variance:σ^2
- Symmetric, bell-shaped

Standardization

$$Z = \frac{X - \mu}{\sigma}$$

Lecture 6 — Empirical Rule & Chi-Square

Empirical Rule

-$P(|Z| < 1) \approx 0.68$
-$P(|Z| < 2) \approx 0.95$
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Chi-Squared Distribution

$$Q = Z_1^2 + \cdots + Z_k^2 \sim \chi_k^2$$

Key Idea

- Sum of squared standard normals

Lecture 7 — Gamma & Beta

Gamma Distribution

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$$

- Models waiting time

Key Relationship

$\text{Gamma}(1, \lambda) = \text{Exponential}$

Beta Distribution

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1$$

- Models proportions

Lecture 8 — Weibull & Log-Normal

Weibull Distribution

$$f(x) = \gamma \lambda x^{\gamma-1} e^{-\lambda x^\gamma}$$

- Flexible lifetime model

Log-Normal Distribution

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Key Idea

- If $\ln X \sim N(\mu, \sigma^2)$, then X is log-normal

Lecture 9 — Summary & Relationships

When to Use

- Uniform → equal likelihood
- Normal → symmetric data
- Exponential → waiting time
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- Beta → proportions
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- Log-normal → positive, skewed data

Key Relationships

- Exponential ↔ Poisson (waiting vs count)
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- Continuous → use integrals
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- CDF accumulates probability
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Module 5 — Moment Generating Functions

Lecture 1 — Moments & MGF

Moments

- k-th moment:

$$E(X^k)$$

- k-th central moment:

$$E[(X-\mu)^k]$$

Skewness

$$\text{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$

Kurtosis

$$\text{Kurt}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] - 3$$

Moment Generating Function (MGF)

$$M_X(t) = E(e^{tX})$$

Key Properties

-$M(0) = 1$

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Lecture 2 — Using MGF

Moments from MGF

- Take derivatives:

$$E(X^k) = M_X^{(k)}(0)$$

First Two Moments

$$E(X) = M_X'(0)$$

$$E(X^2) = M_X''(0)$$

Variance

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Transformation Property

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$$M_Y(t) = e^{at} M_X(bt)$$

Key Idea

- MGF turns calculus → moments

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Sum of Independent RVs

If $U = X_1 + \cdots + X_n$:

$$M_U(t) = M_{X_1}(t) \cdots M_{X_n}(t)$$

Key Insight

- MGFs multiply for independent sums

Important Results

- Sum of normals → normal
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Example Pattern

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- MGF uniquely determines distribution
- Derivatives give moments
- Products handle sums
- Useful for proving distribution results

Module 5 — Moment Generating Functions

Lecture 1 — Moments & MGF

Moments

- k-th moment:

$$E(X^k)$$

- k-th central moment:

$$E[(X-\mu)^k]$$

Skewness

$$\text{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$

Kurtosis

$$\text{Kurt}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] - 3$$

Moment Generating Function (MGF)

$$M_X(t) = E(e^{tX})$$

Key Properties

- $M(0) = 1$
- Encodes all moments
- Must be finite near $t = 0$

Lecture 2 — Using MGF

Moments from MGF

Take derivatives:

$$E(X^k) = M_X^{(k)}(0)$$

First Two Moments

$$E(X) = M_X'(0)$$

$$E(X^2) = M_X''(0)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2$$

Transformation Property

If $Y = a + bX$:

$$M_Y(t) = e^{at} M_X(bt)$$

Key Idea

- MGF turns calculus into moments

Lecture 3 — Sums of Random Variables

Sum of Independent RVs

If $U = X_1 + \cdots + X_n$:

$$M_U(t) = M_{X_1}(t) \cdots M_{X_n}(t)$$

Key Insight

- MGFs multiply for independent sums

Important Results

- Sum of normals $\to$ normal
- Sum of Poissons $\to$ Poisson

Example Pattern

If $X_i \sim \mathrm{Bern}(p)$, then:

$$\sum X_i \sim \mathrm{Binomial}(n, p)$$

Core Takeaways

- MGF uniquely determines distribution
- Derivatives give moments
- Products handle sums
- Useful for proving distribution results

Module 7 — Limit Theorems

Lecture 1 — Inequalities & LLN

Jensen's Inequality

If g is convex:

$$E[g(X)] \geq g(E[X])$$

If g is concave:

$$E[g(X)] \leq g(E[X])$$

Examples:

- $E\left(\frac{1}{X}\right) \geq \frac{1}{E(X)}$ for $X > 0$
- $E[\log X] \leq \log(E[X])$

Markov's Inequality

For $X > 0$, $a > 0$:

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Chebyshev's Inequality

If $E(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Chernoff Inequality

$$P(X \geq a) \leq \frac{E(e^{tX})}{e^{ta}}, \quad t > 0$$

Law of Large Numbers (LLN)

Sample mean:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

Strong LLN

$$P(\bar{X}_n \to \mu) = 1$$

- Converges with probability 1

Weak LLN

For all > 0:

$$P(|\bar{X}_n - \mu| >) \to 0$$

- Probability of large error $\to 0$

Key Idea

- More data $\to$ sample mean $\approx$ true mean

Lecture 2 — CLT

Standardization

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Central Limit Theorem

If X_i are iid with mean μ, variance σ^2:

$$Z_n \xrightarrow{d} N(0, 1)$$

Key Insight

- Regardless of original distribution
- Sample mean becomes approximately normal

Important Approximation

If $Y \sim \mathrm{Bin}(n, p)$ and n large:

$$Y \approx N(np, np(1-p))$$

Takeaway

- CLT = why normal shows up everywhere

Lecture 3 — Normal Approximation & t Distribution

Normal Approximation

Use CLT for probabilities when exact is hard

When CLT Works

- Large n
- Finite mean and variance

Warning

- Does NOT work for distributions like Cauchy
  (no mean or variance)

Sample Variance

$$S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$$

Chi-Squared Result

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}$$

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t Distribution

If:

- $Z \sim N(0,1)$
- $V \sim \chi^2_n$
- independent

Then:

$$T = \frac{Z}{\sqrt{V/n}} \sim t_n$$

Key Properties

- Heavy tails
- Used when σ unknown
- As $n \to \infty$, $t_n \to N(0,1)$

Big Picture

- LLN → convergence
- CLT → shape becomes normal
- t-dist → small sample inference